

A magnitude lower bound and a lattice-clustering reformulation of the Umans–Wang (α, β) -divisor conjecture

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Abstract

The Umans–Wang (α, β) -divisor conjecture (arXiv:2511.10851) asks, for fixed $\alpha, \beta < 1/2$, whether there exist sets $S, T \subseteq \mathbb{Z}^+$ with $|S|, |T| \leq n^\beta$ and elements at most $\exp(n^\alpha)$ such that every $i \in [n]$ divides $s - t$ for some $s \in S, t \in T$; a positive answer would break the exponent-3/2 barrier for polynomial factorization. We record two things. First, an elementary but self-contained *magnitude lower bound* $2\beta + \alpha \geq 1$, obtained via the discriminant of the covering set; this is consistent with the conjecture and reproduces, by an independent route, the bound implied by the primorial. Second, and our main contribution, we reduce the conjecture at fixed α to a precise *lattice-clustering* statement: the covering set is a point of a congruence lattice of covolume $\approx e^{n/2}$ that must lie inside an axis-parallel box of volume e^{Kn^α} , and a generic lattice of that covolume meets the box if and only if $\alpha + \beta \geq 1$. Since the counting floor sits at $\alpha + \beta = (1 + \alpha)/2 < 1$, a solution requires a *special* lattice whose points cluster in an exponentially small box. We show this clustering property is *equivalent* to the conjecture (not weaker), and observe that clustering of this *type* is what the bounded-root-discriminant unit lattices of the 2026 refutation of the real sum–product conjecture by Bloom–Sawin–Schildkraut–Zhelezov provide. We are careful about the degree scaling: our setting appears to require such fields at *polynomial* degree $d \asymp n^{(1-\alpha)/2}$, whereas their construction operates at *logarithmic* degree, so their result does not directly supply what we need; the requisite fields exist at every degree, but whether the shortness carries over at polynomial degree is part of the open question, not something the reduction settles. This isolates a single, frontier-adjacent open question and rules out a family of would-be elementary obstructions.

1 Introduction

Umans and Wang [1] reduce beating the longstanding exponent-3/2 barrier for distinct-degree factorization of degree- n polynomials over $\mathbb{F}_q$ to a number-theoretic conjecture about covering $[n]$ by divisors of a difference set.

Conjecture 1 ((α, β) -divisor conjecture [1]). *For infinitely many n there exist $S, T \subseteq \mathbb{Z}^+$ such that $A = \{s - t : s \in S, t \in T\}$ satisfies the n -divisor property (every $i \in \{1, \dots, n\}$ divides some element of A), with (i) all elements of S, T of magnitude at most $\exp(n^\alpha)$, and (ii) $|S|, |T| \leq n^\beta$.*

Achieving $\alpha + \beta \leq 1 + o(1)$ is trivial; the content is $\alpha, \beta < 1/2$, and $\alpha = \beta = 1/3$ would yield an exponent-4/3 factoring algorithm [1]. Throughout we work at a *fixed* $\alpha < 1/2$ and ask for the smallest feasible β .

Reduction to the primes in $(n/2, n]$. The n -divisor property is governed by its hardest instances, the primes $p \in P(n) := \{p \text{ prime} : n/2 < p \leq n\}$: any $i \leq n$ has all prime-power factors at most n , and the primes in $(n/2, n]$ are those that can divide at most one difference of magnitude $O(n)$ and cannot share factors. We therefore study covering $P(n)$ by divisibility. By the prime number theorem $\log \prod_{p \in P(n)} p = \theta(n) - \theta(n/2) = \frac{n}{2}(1 + o(1))$ (verified numerically: the ratio is 0.963, 0.997, 0.999 at $n = 10^3, 10^4, 10^5$).

2 The magnitude lower bound

We first give a clean lower bound. Write $M = \exp(n^\alpha)$ and, for a covering set S with $|S| = K = n^\beta$, let $D(S) = \prod_{i < j} (s_i - s_j)$ be the Vandermonde product (the square root of the discriminant of $\prod_i (x - s_i)$).

Lemma 2. *If the difference set of S covers $P(n)$ by divisibility, then $Q := \prod_{p \in P(n)} p$ divides $D(S)$.*

Proof. Each $p \in P(n)$ divides some difference $s_i - s_j$, hence divides $D(S)$. The primes of $P(n)$ are distinct, so their product Q divides $D(S)$. $\square$

We verified Lemma 2 directly on explicit covers at $n = 20, 30, 40$ (e.g. $S = \{0, 17, 20, 39\}$ covers $P(20)$ and $Q = 46189 \mid D(S)$).

Theorem 3. *Any S, T satisfying Conjecture 1 obey $2\beta + \alpha \geq 1 - o(1)$.*

Proof. Take S with $|S| = K = n^\beta$ (the symmetric case; the asymmetric $S - T$ case is identical with K^2 in place of $\binom{K}{2}$). By Lemma 2, $Q \mid D(S)$, so $\log Q \leq \log |D(S)|$. Each of the $\binom{K}{2}$ factors of $D(S)$ has magnitude at most $2M$, hence

$$\frac{n}{2}(1 - o(1)) = \log Q \leq \binom{K}{2} \log(2M) \leq \frac{K^2}{2} n^\alpha.$$

Thus $K^2 n^\alpha \geq n(1 - o(1))$, i.e. $2\beta + \alpha \geq 1 - o(1)$. $\square$

Remark 4. Theorem 3 is consistent with Conjecture 1: at $\alpha = \beta = 0.4$ one has $2\beta + \alpha = 1.2 \geq 1$. It reproduces, via the discriminant, the bound that Umans–Wang obtain from $\prod_{p < n} p = e^{n(1+o(1))}$. It is tight *as a magnitude bound*: it cannot by itself yield a stronger exponent, and in particular cannot decide the conjecture. The boundary $2\beta + \alpha = 1$ is the *counting floor* $\beta = (1 - \alpha)/2$.

3 The lattice-clustering reformulation

We now reduce the conjecture at fixed α to a single geometric statement. Fix $\alpha < 1/2$ and set $K = n^{(1-\alpha)/2}$ (the counting floor), $B = \exp(n^\alpha)$.

The congruence lattice. A covering assignment attaches to each prime $p \in P(n)$ a pair of indices whose difference p must divide. Collecting these, the admissible tuples $s = (s_1, \dots, s_K)$ form a coset structure inside the lattice

$$L = \{s \in \mathbb{Z}^K : s_i \equiv s_j \pmod{m_{ij}} \text{ for each assigned pair}\},$$

where m_{ij} is the product of primes assigned to pair (i, j) . Since each prime of $P(n)$ is assigned to exactly one pair, the moduli are pairwise coprime and the index $[\mathbb{Z}^K : L] = \prod_{p \in P(n)} p = Q$, so L has covolume $Q = e^{n/2(1+o(1))}$ in $\mathbb{R}^K$.

The box condition. A covering set of magnitude $\leq B$ with distinct coordinates is exactly a point of L in the box $[0, B]^K$ with distinct coordinates. The box has volume $B^K = \exp(Kn^\alpha)$. Comparing to the covolume:

$$\frac{\text{vol}[0, B]^K}{\text{covol}(L)} = \exp\left(Kn^\alpha - \frac{n}{2}\right), \quad Kn^\alpha = n^{(1-\alpha)/2} \cdot n^\alpha = n^{(1+\alpha)/2}.$$

Proposition 5. *For a generic lattice of covolume Q , the expected number of points in $[0, B]^K$ is $B^K/Q = \exp(n^{(1+\alpha)/2} - \frac{n}{2})$. This is ≥ 1 if and only if $Kn^\alpha \geq \frac{n}{2}$, i.e. (writing $K = n^\beta$) if and only if $\alpha + \beta \geq 1$.*

At the counting floor $\beta = (1 - \alpha)/2$ we have $\alpha + \beta = (1 + \alpha)/2 < 1$, so the box is exponentially smaller than the covolume and a *generic* congruence lattice contains *no* admissible point (numerically, $B^K/Q = e^{-\Theta(n)}$; the ratio $Kn^\alpha/\log Q$ is 0.017 and 0.032 at $n = 10^6$, $\alpha = 0.3, 0.4$). Hence:

Theorem 6 (Reformulation). *Conjecture 1 at fixed α is equivalent to the existence of congruence lattices L as above whose points cluster: L must contain a distinct-coordinate point in a box that is a factor $e^{\Theta(n)}$ smaller in volume than $\text{covol}(L)$. Equivalently, L must beat the generic point-count by an $e^{\Theta(n)}$ factor.*

Proof sketch. A solution to Conjecture 1 at α (with β at the floor) is, after fixing an assignment of primes to pairs, precisely a distinct-coordinate point of the corresponding L in $[0, B]^K$; conversely such a point yields a covering set of the required magnitude and cardinality. The magnitude/cardinality bookkeeping is Theorem 3 and Proposition 5. The freedom to choose the assignment does not change the covolume Q , only the shape of L ; so feasibility is exactly the existence of *some* assignment whose lattice clusters into the box. $\square$

Remark 7 (The reduction does not simplify the problem, and that is the point). Theorem 6 is an *equivalence*, not a weakening. Any attempt to prove the clustering as a lighter sub-lemma is therefore misguided: it carries the full weight of the conjecture. The value of the reformulation is the change of language—from divisor covering to lattice clustering—which connects the problem to a concrete construction technology (below) and, as we discuss in the companion note, explains why an entire family of elementary “barrier” arguments fail: they attempt to lower-bound coordinates of L by a generic (Minkowski-average) estimate, which the clustering hypothesis is precisely designed to beat.

4 The construction frontier

The clustering property in Theorem 6 is not abstract: it is exactly the property exploited by Bloom, Sawin, Schildkraut and Zhelezov [2] in their 2026 refutation of the real sum–product conjecture. They construct sets $A = GP$ in totally real number fields K of degree $d \asymp \log |A|$ with bounded root discriminant (Martinet towers), where G is a box in the unit lattice (multiplicatively structured, small product set) and P a box of algebraic integers (additively structured). In such a field the unit lattice has $O(1)$ -size-per-dimension vectors despite large covolume—exactly the super-generic clustering our reformulation requires.

We record the quantitative situation, and flag honestly where it is favorable and where it is not. To place $K = n^{(1-\alpha)/2}$ coordinates below $B = e^{n^\alpha}$ against covolume $Q = e^{n/2}$, one needs shortness beating the Minkowski average $Q^{1/K} = e^{n^{(1+\alpha)/2}/2}$ by a factor growing polynomially in n

(numerically $9\times, 58\times, 316\times$ at $\alpha = 0.3$, $n = 10^4, 10^6, 10^8$). A degree- d bounded-root-discriminant field has unit lattice of rank $d - 1$ with $O(1)$ -size-per-dimension vectors, which is shortness of the required *type*; and Martinet towers provide bounded-root-discriminant fields at *every* degree d , so in particular the fields exist at any degree we might ask for.

However, the degree scaling is a genuine soft point of this reduction, and we do not claim it is resolved. Placing K coordinates subject to K constraints calls, naively, for on the order of K independent short directions, hence degree $d \asymp K = n^{(1-\alpha)/2}$ —*polynomial* in n . This is a different regime from the construction of [2], whose degree scales *logarithmically*, $d \asymp \log |A|$, in the size of the set built; the ratio $K/\log n$ grows without bound (it exceeds 500 already at $\alpha = 0.3$, $n = 10^{12}$). Thus one cannot simply cite [2] for the shortness we need: their $O(1)$ -per-dimension bound is established for sets of size $|A|$ at degree $\log |A|$, whereas using degree $d \asymp K$ corresponds to an object of size $|A| \asymp e^K$, larger than and distinct from what they analyze. What the Martinet input gives us is the *existence* of the fields at the required (polynomial) degree; whether their unit lattices supply K *usable, independent* short directions respecting our specific pair-congruences is not delivered by [2] and is, in fact, part of the open question below.

Consequently the open question is twofold and precise: whether the specific pair-congruences of L can be embedded into such a unit lattice (i) *at the required polynomial degree* $d \asymp K$, and (ii) *preserving $O(1)$ -per-dimension shortness*, so that a simultaneous solution descends to K distinct integers $\leq B$. This is a *descent-to- $\mathbb{Z}$* question of the type [2] leave open (they resolve the $\mathbb{R}$, p -adic, and finite-field cases; the integer case, and hence the sum-product problem in $\mathbb{Z}$, remains open), now with the additional, unresolved demand that the shortness survive at polynomial rather than logarithmic degree.

Structural notes for a specialist. Three facts constrain any attack. (1) The conjecture uses an *asymmetric* difference set $S - T$; the two-set freedom is genuine (exact search shows it beats the naive per-slot magnitude bound by a growing factor, verified at $K \leq 3$), and it matches the $G \cdot P$ split of [2], with S playing the role of the multiplicative box and T the additive box. (2) The relevant regime is *bundled*: at the floor each difference must be divisible by $\Theta(n^\alpha/\log n)$ primes of $P(n)$ simultaneously, forcing magnitude $\approx e^{n^\alpha}$; small-magnitude or polynomial-magnitude computation cannot probe it (a difference $\leq 2n$ carries at most one prime $> n/2$). (3) A solution must be *structured* (efficiently computable) for the algorithmic application, so an existence proof alone does not suffice; the explicit [2]-style construction is the right kind of target.

5 Conclusion

We have isolated the Umans–Wang conjecture at fixed α as a lattice-clustering problem equivalent to the conjecture itself, and tied it to the concrete bounded-root-discriminant construction technology of [2]. The magnitude bookkeeping is favorable and the requisite fields exist at every degree; but the degree scaling is not resolved—our setting appears to demand shortness at *polynomial* degree $d \asymp n^{(1-\alpha)/2}$, while [2] supply it at *logarithmic* degree—so the reduction leaves a descent-to- $\mathbb{Z}$ embedding question that is genuinely open and includes this degree gap, rather than a merely technical residue. The proven bound $2\beta + \alpha \geq 1$ is the only unconditional statement; the conjecture remains open in both directions. What we can claim is narrower but solid: the elementary obstructions to the conjecture provably fail (they are artifacts of generic estimates), and the natural constructive route is now explicit and precisely delimited, including an honest account of where it is not yet supported.

References

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